

2026 年 01 月 CSCA 考试物理试题

Physics January 2026 CSCA

Instructions

Each question has four options among which only one of them is correct.

1. The 26th International Conference on Weights and Measures decided to redefine the basic unit of mass, which came into effect on May 20, 2019. The basic unit of mass is
 - (A) N
 - (B) s
 - (C) m
 - (D) kg
2. The gravitational acceleration on the surface of the moon is about $\frac{1}{6}$ of that on Earth. If the gravity of an object on Earth is 120N, its gravity on Moon is
 - (A) 60N
 - (B) 20N
 - (C) 120N
 - (D) 720N
3. The natural length of a spring is 0.7m, and its spring constant is $k = 10, \text{N/m}$. When a force of 3N is applied to stretch the spring, its new length is
 - (A) 0.5m
 - (B) 0.7m
 - (C) 0.3m
 - (D) 1.0m
4. Suppose that an object is subjected to two forces F_1 and F_2 . Given that $F_1 = 3, \text{N}$ and $F_2 = 5, \text{N}$, the magnitude of the resultant force of these two forces cannot be

(A) 2N

(B) 9N

(C) 5N

(D) 8N

5. In a uniform magnetic field, a straight wire with length 0.5m is placed perpendicular to the direction of the magnetic field. When passing through a current of 10A it experiences an Ampere force of 1.5N. The magnitude of the magnetic induction intensity of the magnetic field is

(A) 0.3T

(B) 0.2T

(C) 0.4T

(D) 0.1T

6. Table tennis is very suitable for the characteristics of Asians and is the national sport of China. Which of the following statements about table tennis is correct?

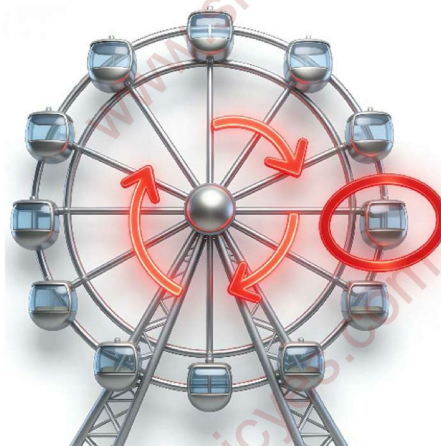
(A) Ping Pong is relatively small and can be viewed as a particle at any time

(B) When studying the flight path of the ball, it can be viewed as a particle

(C) The displacement of the ball during its flight in the air is equal to the distance traveled

(D) When studying the rotation of the ball, one must consider it as a particle

7. The radius of a Ferris wheel is R . The cabin rotates with the wheel once, and its distance and displacement are



(A) distance $2\pi R$, displacement $2R$

(B) distance $2R$, displacement $2\pi R$

(C) distance $2R$, displacement 0

(D) distance $2\pi R$, displacement 0

8. Which of the following statements is incorrect?

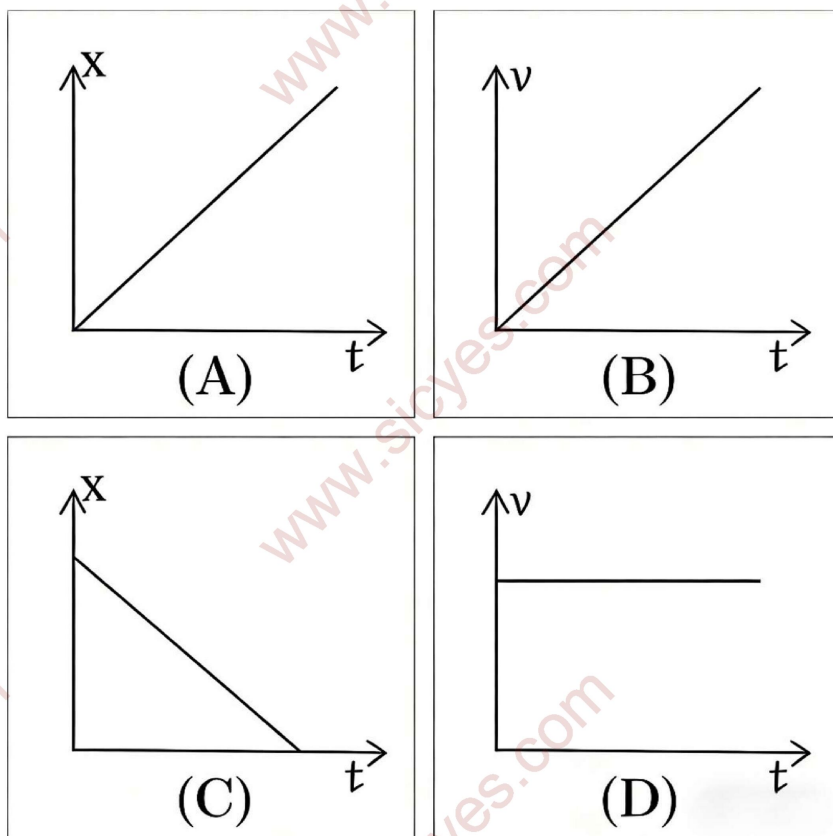
(A) Sprinkling some fine soil on a frozen horizontal road surface would make it less likely for people to slip on it, because the maximum static friction between the people and the road surface increases

(B) Hold a bottle in the hand vertically in the air. When the grip of the hand is increased, the frictional force on the bottle remains unchanged

(C) When walking, people are subjected to static friction

(D) When a person climbs a vertically fixed bamboo pole with both hands at a constant speed, the direction of the frictional force experienced is downward

9. Which of the following graphs represents the motion of a car moving on a straight line with constant acceleration?



10. An object falls freely from a height of 5m. The time it takes to reach the ground is ($g = 10, \text{m/s}^2$)

(A) 4s

(B) 1s

(C) 2s

(D) 3s

11. An object moves in a uniform circular motion with a radius of 0.1m and a period of 5s. The linear velocity and angular velocity of the object during motion are, respectively,

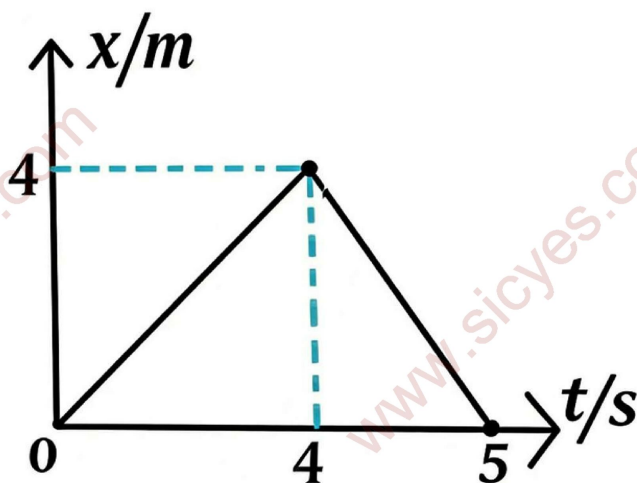
(A) $100\pi, \text{m/s}$, $0.4\pi, \text{rad/s}$

(B) $0.4\pi, \text{m/s}$, $0.4\pi, \text{rad/s}$

(C) $0.5\pi, \text{m/s}$, $100\pi, \text{rad/s}$

(D) $0.04\pi, \text{m/s}$, $0.4\pi, \text{rad/s}$

12. The graph of the displacement (x) of a toy car moving in a straight line changing with time (t) is shown in the figure. Which of the following statements is correct?



(A) Within 0-4 seconds, the displacement of the toy car's movement is 8m

(B) Within 0-4 seconds, the speed of the toy car is 1.0m/s

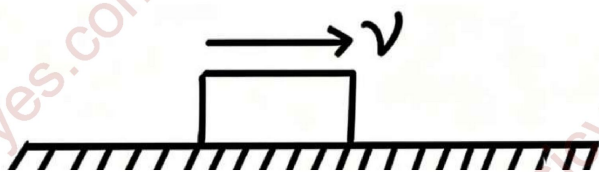
(C) Within 0-5 seconds, the average speed of the toy car is 0.8m/s

(D) Within 4-5 seconds, the speed of the toy car is 0.8m/s

13. When an elevator accelerates vertically upwards with an acceleration of $2, \text{m/s}^2$, the pressure exerted on the floor by a person with a mass of 50kg inside the elevator is ($g = 10, \text{m/s}^2$)

(A) 500N
(B) 300N
(C) 400N
(D) 600N

14. As shown in the figure, an object with a mass of 1kg slides to the right along a horizontal ground. Given that the sliding friction force acting on the object is 3N, and $g = 10, \text{m/s}^2$, the coefficient of kinetic friction between the object and the horizontal ground is



(A) 0.2
(B) 0.3
(C) 1
(D) 0.4

15. On a smooth horizontal surface, a constant force F is applied horizontally to a stationary object with a mass of 1.0kg, and the magnitude of its velocity change within 1 second is measured to be 10m/s. Therefore, the magnitude of the force F is

(A) 40N
(B) 5N
(C) 10N
(D) 20N

16. When light enters a medium from air, with an angle of incidence $i = 60^\circ$, the

angle of refraction is $r = 30^\circ$. If the refractive index of air is $n = 1$, the refractive index of the medium is

(A) $\sqrt{3}$

(B) 3

(C) 1.5

(D) $\sqrt{5}$

17. A small ball is in uniform circular motion with an angular velocity of 100π rad/s. The distance between the ball and the center of the circle is 1cm. The centripetal acceleration of the ball is

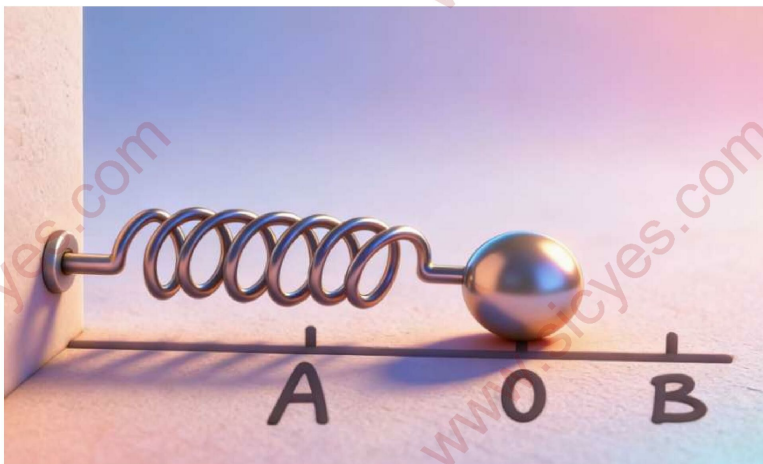
(A) $50\pi^2$, m/s²

(B) $100\pi^2$, m/s²

(C) $10000\pi^2$, m/s²

(D) π , m/s²

18. As shown in the figure, a spring oscillator performs a simple harmonic motion between A and B, with O being the equilibrium position. The distance between A and B is measured to be 12cm, and the ball completes 5 full oscillations in 60 seconds. Thus,



(A) If the timing starts when the ball passes point O, the distance traveled within 3 seconds is 24cm

(B) The ball travels 12cm in one complete oscillation

(C) The period of the oscillator is 2s, with an amplitude of 6cm

(D) The frequency of the oscillator is 2Hz

19. An object with a mass of 50kg moves horizontally towards the right at a speed of 10m/s. The momentum and kinetic energy are, respectively,

(A) **500**, kg·m/s , 2500J

(B) **2500**, kg·m/s , 500J

(C) **500**, kg·m/s , 5000J

(D) **5**, kg·m/s , 500J

20. If a water droplet with zero initial velocity falls from a 5-meter high eaves (excluding air resistance), the speed at which the water droplet falls to the ground is ($g = 10, \text{m/s}^2$)

(A) 5m/s

(B) 10m/s

(C) 2m/s

(D) 20m/s

21. A wooden box is pulled by a force $F = 50, \text{N}$ at an angle of 37° to the horizontal direction, which causes it to move a distance of 10m on a horizontal surface. Given that $\cos 37^\circ \approx 0.8$, the work done by the force F on the box is

(A) 400J

(B) 300J

(C) 500J

(D) 250J

22. A simple harmonic transverse wave propagates along the positive direction of the x-axis, with a wave velocity of 10m/s. The waveform at $t = 0$ is shown in the

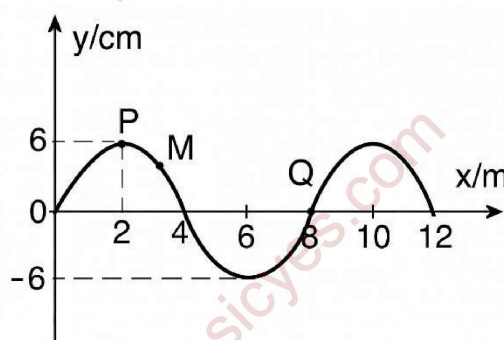


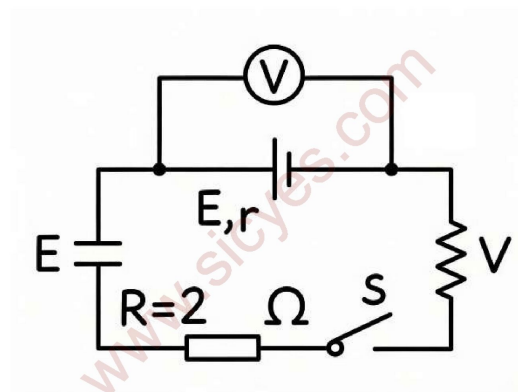
figure. Which of the following statements is correct?

- (A) The wavelength is 12m, the period is 1.2s
- (B) The wavelength is 4m, the period is 0.4s
- (C) Within 0~0.6s, the particle P has travelled 18cm
- (D) The wavelength is 8m, the period is 0.8s

23. Two electrical resistors have resistances 3Ω and 6Ω respectively. Which of the following statements is correct?

- (A) When connected in series, the total resistance is 2Ω ; when connected in parallel, the total resistance is 2Ω
- (B) When connected in series, the total resistance is 9Ω ; when connected in parallel, the total resistance is 9Ω
- (C) When connected in series, the total resistance is 9Ω ; when connected in parallel, the total resistance is 2Ω
- (D) When connected in series, the total resistance is 2Ω ; when connected in parallel, the total resistance is 9Ω

24. In the circuit shown in the figure, the resistor has $R = 2\Omega$. After disconnecting S, the voltmeter reading is 3V; after closing S, the voltmeter reading is 2V. So the internal resistance r of the power supply is



- (A) 3Ω
- (B) 2Ω
- (C) 4Ω
- (D) 1Ω

25. A balloon is filled with ideal gas. When the balloon rises, assume that the volume of the gas remains constant, and the temperature decreases. In this process the ideal gas in the balloon would

- (A) Absorb heat
- (B) Increase pressure
- (C) Decrease internal energy
- (D) Do negative work to outside

26. During the ascent of an object being thrown vertically upwards, gravity does -70 J of work on the object. Ignoring air resistance, which of the following statements is correct?

- (A) The gravitational potential energy decreases by 70 J
- (B) The gravitational potential energy increases by 70 J
- (C) The mechanical energy decreases by 70 J
- (D) The gravitational potential energy remains constant

27. If the current passing through a wire is 1.6 A , the amount of charge passing through the cross-section of the wire within 0.5 s is

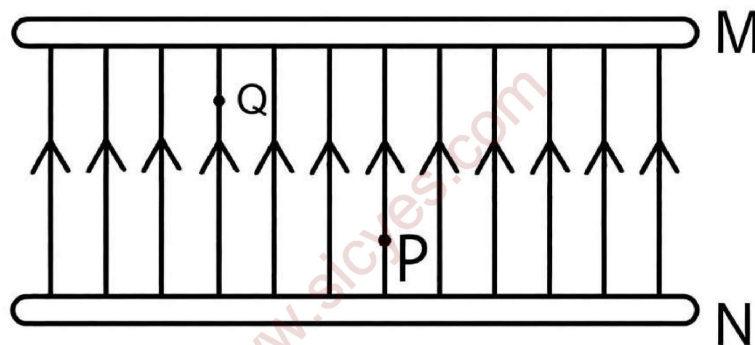
- (A) 1.6 C
- (B) 1.2 C
- (C) 2.0 C
- (D) 0.8 C

28. A certain mass of ideal gas, with its temperature kept constant and pressure reduced, its volume would

- (A) All options above are possible
- (B) Increase
- (C) Decrease
- (D) Remain unchanged

29. Three point charges A, B, and C are located at the vertices of an equilateral

30. As shown in the figure, two parallel metal plates M and N with equal and opposite charges are placed horizontally. The electric field lines between them are parallel lines with equal intervals, and P and Q are two points in the electric field. Which of the following statements is correct?



- (A) M has positive charge, N has negative charge
- (B) P, Q have equal field strength
- (C) Both the electrical potential and field strength at P, Q are different
- (D) P, Q have equal electrical potential

31. A straight wire with length $L = 0.5\text{ m}$ and a current of $I = 2\text{ A}$ is placed in a uniform magnetic field with a magnetic induction intensity of $B = 0.4\text{ T}$. The angle between the direction of the magnetic field and the direction of the current in the wire is 30 degrees. The ampere force acting on the wire is

(A) 0.4 N

(B) 0 N

(C) 0.2 N

(D) $\frac{\sqrt{3}}{5}\text{ N}$

32. In a uniform magnetic field with a magnetic induction intensity of $B = 0.8\text{ T}$, a rectangular coil has area $S = 0.5\text{ m}^2$, and the angle between the normal direction of the coil plane and the direction of the magnetic field is 60 degrees. Therefore, the magnitude of the magnetic flux passing through the coil is

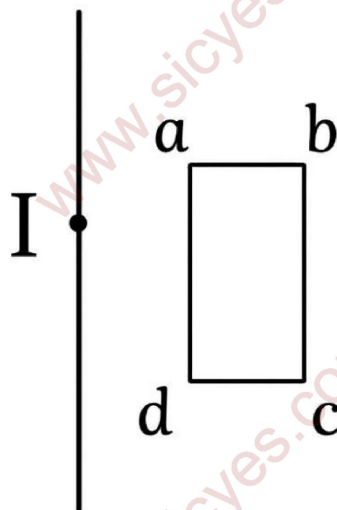
(A) 0.2 Wb

(B) 0.8 Wb

(C) 0.4 Wb

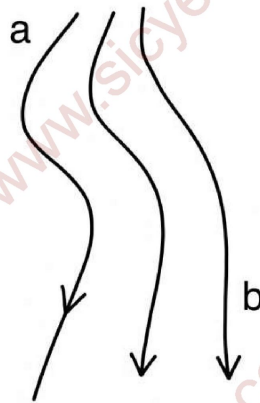
(D) 0.6 Wb

33. As shown in the figure, on the right side of a long straight wire with a current I flowing in the direction indicated in the diagram, a rectangular metal wire frame $abcd$ is fixed, with its side cd parallel to the wire. By adjusting the current I , the magnetic induction intensity at each point in space increases uniformly over time. Then,



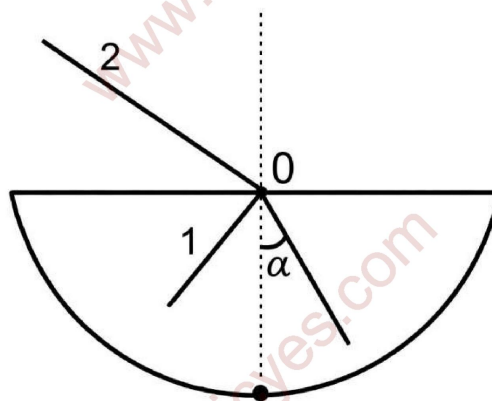
- (A) The induced current in the frame has direction $a \rightarrow b \rightarrow c \rightarrow d \rightarrow a$
- (B) The induced current in the frame gradually increases
- (C) The frame experiences an ampere force horizontally towards the right
- (D) The ampere force acting on the edge cd is constant

34. As shown in the figure, there is an electric field in space with two points a and b. Which of the following statements is correct?



- (A) The electric field strength at point b is greater than that at point a
- (B) The electric field strength at point a is greater than that at point b
- (C) This electric field is a uniform electric field
- (D) The electric field strength at points a and b is the same

35. As shown in the figure, the cross-section of a glass cylinder is semi-circular. A thin monochromatic beam travels through the cylinder towards the center of the semicircle O at an incident angle of $\alpha = 30^\circ$. The reflected beam 1 and refracted beam 2 produced are exactly perpendicular to each other. Therefore, the refractive index of the glass is



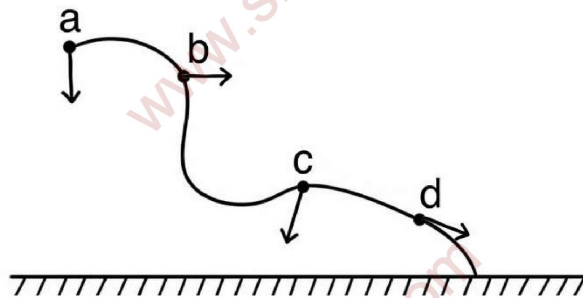
(A) $\sqrt{3}$

(B) $\sqrt{2}$

(C) $\frac{1}{\sqrt{3}}$

(D) $\frac{\sqrt{3}}{2}$

36. The trajectory of a leaf falling from the air is shown in the figure, the velocity directions of the leaf at positions a, b, c, and d are indicated. The correct direction is



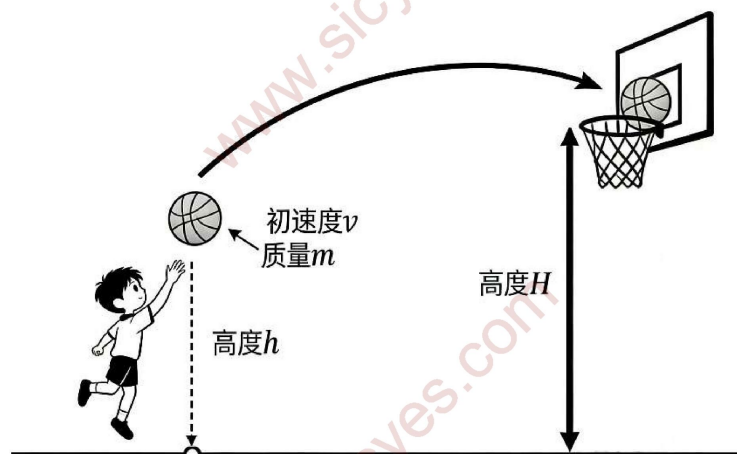
(A) b

(B) c

(C) a

(D) d

37. As shown in the figure, a student throws a basketball with a mass of m from a height of h at an initial velocity of v , and the basketball just enters the basket with a height of H above the ground. Taking the ground as the zero potential energy surface, ignoring air resistance, and let the magnitude of gravitational acceleration be g , the mechanical energy of the basketball entering the basket is



(A) $\frac{1}{2}mv^2 + mgH$

(B) $\frac{1}{2}mv^2 + mg(H - h)$

(C) $\frac{1}{2}mv^2 + mgh$

(D) $\frac{1}{2}mv^2 + mgH + mg(H - h)$

38. A car with a mass of 2 tons and an engine with a power of 30kW travels at a constant velocity of 72km/h on a level ground. The resistance experienced by the car while driving on that road is

(A) $5 \times 10^3, \text{N}$

(B) $6 \times 10^3, \text{N}$

(C) $1.5 \times 10^3, \text{N}$

(D) $2 \times 10^3, \text{N}$

39. A ball is thrown horizontally from a certain height at an initial velocity of v_0 and it lands on a level ground. Then, the ball is thrown again at the same height horizontally at an initial velocity of $2v_0$ and it lands on the same ground. Ignoring air resistance, during the second throw of the ball,

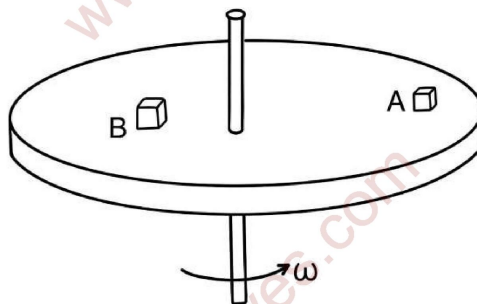
(A) The horizontal displacement becomes twice of the original

(B) The time taken to fall becomes twice of the original

(C) The horizontal displacement becomes half of the original

(D) The time taken to fall becomes half of the original

40. As shown in the figure, if two objects A and B with equal mass are placed on a horizontal disk rotating uniformly around the vertical axis, and the two objects remain stationary relative to the disk, then

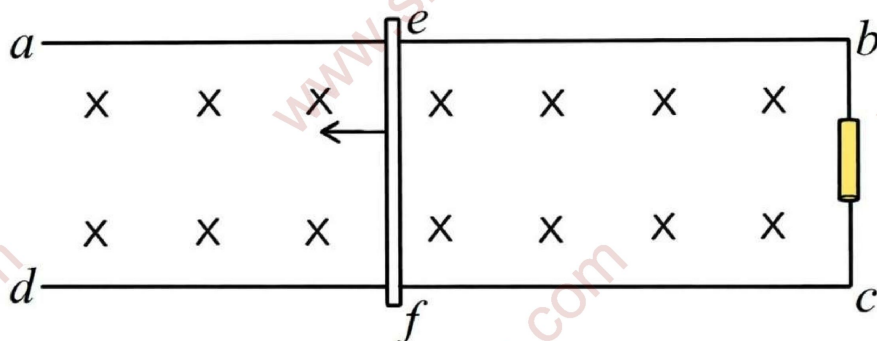


- (A) The angular velocity of A is the same as B
- (B) The linear velocity of A is the same as B
- (C) The angular velocity of A is greater than that of B
- (D) The angular velocity of A is smaller than that of B

41. An object with mass m falls freely from rest. Ignoring air resistance, the power delivered by gravity on the object over time t is

- (A) $\frac{1}{2}mg^2t$
- (B) mgt
- (C) mg^2t
- (D) $\frac{1}{4}mgt$

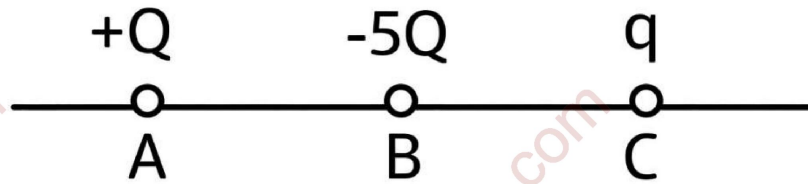
42. As shown in the figure, a fixed U-shaped metallic rail abd with a load connected is placed in a uniform magnetic field perpendicular to the plane of the rails. There is a metal rod ef that can slide freely on the rails. During the process in which the rod ef moves to the left, which of the following statements is correct?



- (A) There is a current flowing through segment ae
- (B) The magnetic flux passing through $ebcf$ remains unchanged
- (C) There is a current flowing through segment bc
- (D) The magnetic flux passing through $ebcf$ decreases

43. As shown in the figure, two identical metal balls (both of which can be considered as point charges) are fixed at points A and B in vacuum, with charges of $+Q$ and $-5Q$, respectively. A point charge q is fixed at point C on the line extending from A to B. The magnitude of the electrostatic force acting

on this charge is F_1 . It is given that $AB = BC$. If the two charged metal balls are brought into contact and then returned to their original positions, the magnitude of the electrostatic force acting on charge q becomes F_2 . Thus, $\frac{F_1}{F_2}$ is



(A) $\frac{21}{10}$

(B) $\frac{19}{10}$

(C) $\frac{21}{16}$

(D) $\frac{19}{16}$

44. A basketball with a mass of $m = 0.6$ kg moves horizontally to the left at a speed of $v_1 = 2$ m/s. When it hits a hard wall, it bounces back horizontally to the right at a speed of $v_2 = 1$ m/s. If the ball is in contact with the wall for 0.1 seconds, the average horizontal force exerted on the basketball by the wall is (Neglect air resistance, $g = 10$ m/s²)

(A) 1.8 N

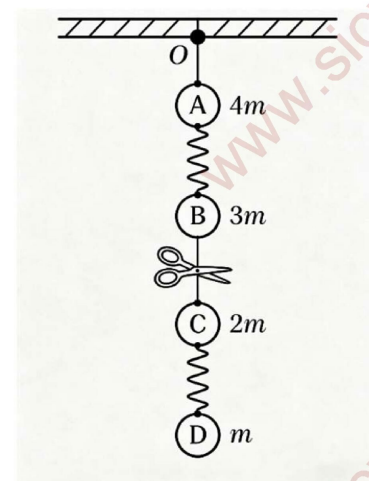
(B) 12 N

(C) 18 N

(D) 6 N

45. As shown in the figure, four small balls A, B, C, and D with masses of $4m$, $3m$, $2m$, and m , respectively, are connected to each other through thin wires or light springs, suspended at point O, in a stationary state with a gravitational acceleration of g . If the thin wire between B and C is cut, the accelerations of B and C at the instant the wire is cut are, respectively:

(A) $2g$, $1.5g$

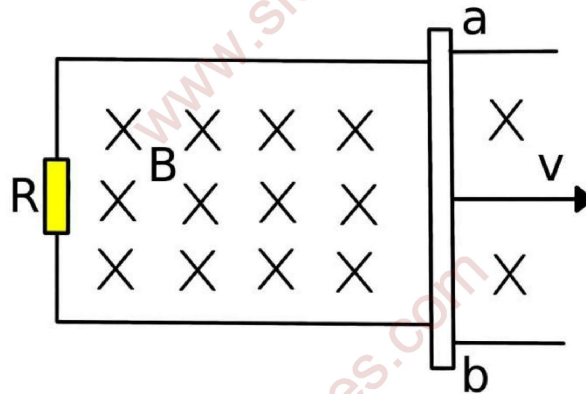


(B) $g, 1.5g$

(C) $g, 0.5g$

(D) $2g, 0.5g$

46. As shown in the figure, the two ends of parallel metal rails placed horizontally are connected with a resistor R . The wire ab can slide on the frame without friction, and a uniform magnetic field passes vertically through the frame plane. When ab moves uniformly to the right, which of the following statements is correct?



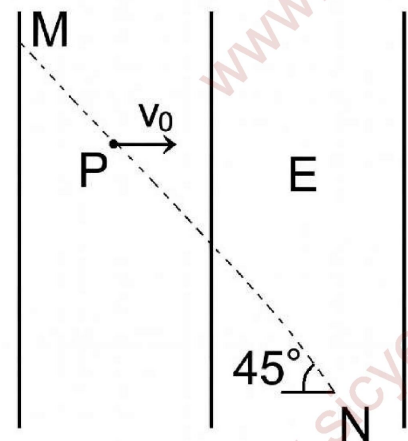
(A) As long as the wire ab reaches a certain velocity and the external force is removed, the wire ab can also maintain a uniform motion on the frame

(B) Wire ab is not only subjected to tension force, but also to magnetic field force

(C) The movement velocity of wire ab is constant. If the resistance value R is increased, the force pulling wire ab can be increased

(D) The higher the velocity of wire ab 's movement, the smaller the required pulling force

47. As shown in the figure, a particle of mass m and charge $q > 0$ is projected horizontally into a uniform electric field of magnitude E directed vertically downward. It is projected from point P on the line MN with an initial velocity v_0 . Given that line MN makes an angle of 45° with the horizontal, and assuming that gravity is negligible, when the particle returns to the line MN :



(A) The angle that the velocity makes with the vertical direction is 30°

(B) The distance from P is $\frac{2\sqrt{2}mv_0^2}{qE}$

(C) The time it takes is $\frac{mv_0}{qE}$

(D) The velocity is $3v_0$

48. It is important to maintain a safe driving distance at high speeds. The maximum deceleration of a small car is 6 m/s^2 . A driver is driving a car on a highway at a speed of 108 km/h . If the driver's reaction time is 1.5 seconds, the minimum safe distance from the car ahead should be at least

(A) 90 m

(B) 60 m

(C) 120 m

(D) 150 m